UI Design Using RAD Model for RapidAid: Smart Emergency Ambulance & Hospital Recommendation System

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AIM:

To simulate the lifecycle stages of UI design using the RAD (Rapid Application Development) model and develop an interactive emergency healthcare mobile application prototype named **RapidAid**.

INTRODUCTION:

RAD (Rapid Application Development) is a software development methodology that emphasizes rapid prototyping, iterative development, and continuous user feedback.

In this experiment, an emergency-based mobile application named **RapidAid** is designed. The application helps users quickly request emergency ambulance services, track real-time ambulance location, and get nearby hospital recommendations. The system focuses on speed, usability, and life-saving efficiency.

PHASE 1: REQUIREMENTS PLANNING

1. Identify Key Features

Navigation:

- Welcome / Login Screen
- Home Dashboard
- Emergency Request Screen
- Emergency Confirmation Screen
- Live Tracking Screen
- Hospital Recommendation Screen
- Profile & Medical History Screen

User Actions:

- Login / Register
- Request emergency help
- Select emergency type (Cardiac, Accident, etc.)
- Confirm emergency request
- Track ambulance in real-time
- View nearby hospitals
- View personal medical details

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2. Functional Requirements

1. User can log in to the application
2. User can request emergency help instantly
3. System identifies emergency type
4. System assigns ambulance automatically
5. User can track ambulance location live
6. System suggests nearby hospitals
7. User can view emergency status updates
8. User can access profile and medical records

PHASE 2: USER DESIGN

Tool Used:

Figma

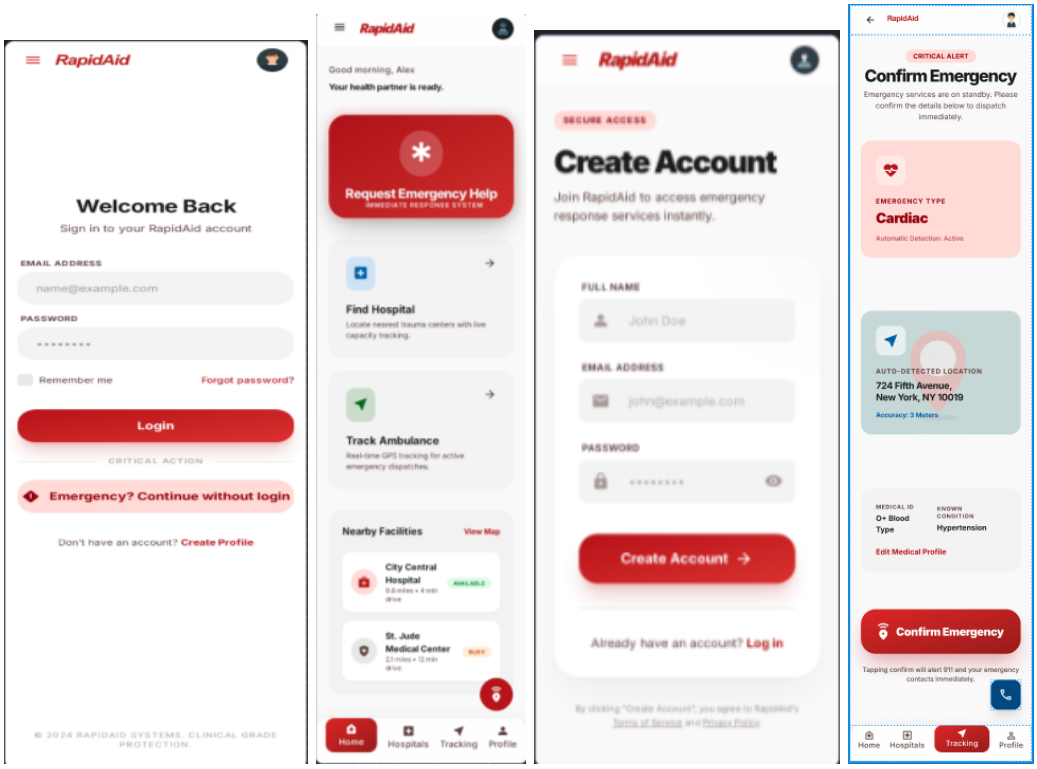
Wireframes / Screens Designed:

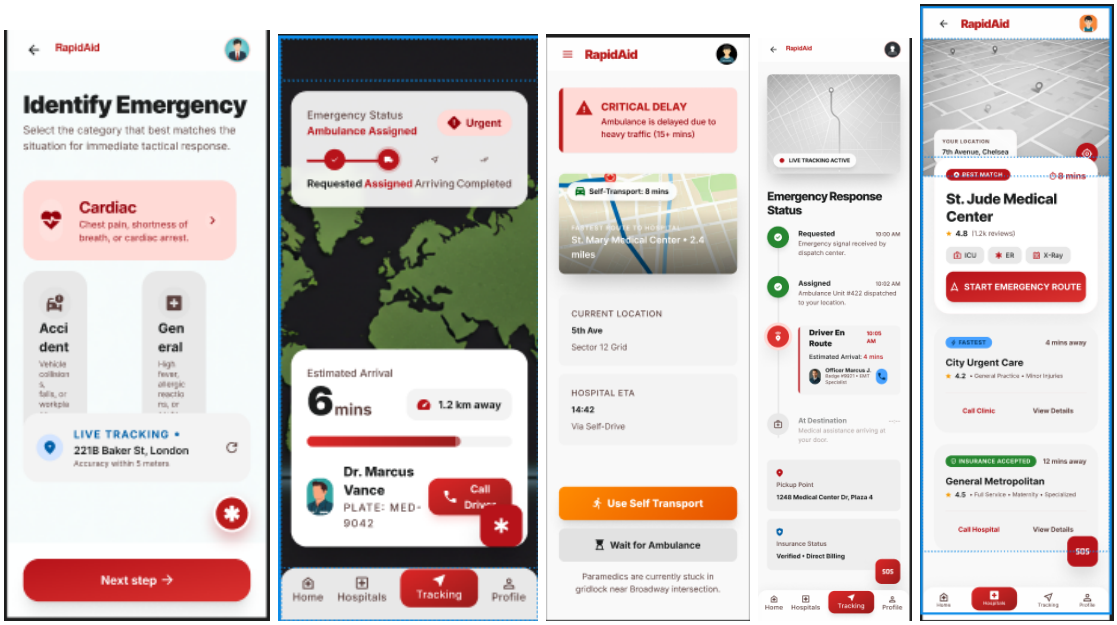
- Login & Welcome Screen
- Home Dashboard with Emergency Button
- Emergency Type Selection Screen
- Emergency Confirmation Screen
- Live Tracking & Status Screen
- Hospital Recommendation Screen
- Profile & Emergency History Screen

Interactions Implemented:

- Button click navigation between screens
- Emergency request trigger button
- Real-time tracking UI simulation
- Status updates (Requested → Assigned → Arrived)
- Smooth transitions between screens
- Interactive cards for hospital selection

SCREENSHOTS





PHASE 3: CONSTRUCTION

1. Interactive Prototype

- All screens are connected using Figma prototype links
- Navigation flow implemented:

Login → Home → Emergency Request → Confirmation → Tracking → Hospital Selection → Profile

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2. Dynamic Features

- Emergency alert button with highlight effect
- Status indicators (color-based: red, yellow, green)
- Live tracking map interface simulation
- Smooth UI transitions
- Card-based hospital UI

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3. Testing

- Prototype tested using Figma preview mode
- Verified smooth navigation between screens
- Improved UI clarity and accessibility
- Ensured fast emergency flow usability

PHASE 4: CUTOVER

1. Finalization

- Completed UI prototype of RapidAid
- Ready for real-time implementation

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2. Documentation

- All UI screens documented
- Emergency flow clearly defined
- Interaction behavior explained

RESULT:

Thus, the emergency healthcare mobile application **RapidAid** was successfully designed using Figma following the RAD model. The prototype demonstrates a fast, efficient, and user-friendly emergency response system with real-time tracking and hospital recommendation features.